The University of Texas at Austin
In-Class Work 18

M339D Introduction to Financial Mathematics

November 09, 2026

Instructions: Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

Log-Normal Stock Prices: Tail Probabilities.

Motivation.

Problem 18.1: *Let the stock prices be modeled using the lognormal distribution. Under the risk-neutral probability measure, the mean stock price at time*-1 *equals* 120 *and the median stock price* 115*. What is the risk-neutral probability that the time*-1 *stock price exceeds* 100*?*

Derivation.

Problem 18.2: *Consider a non-dividend-paying stock. What is the probability that the stock outperforms a risk-free investment* under the risk-neutral probability measure $\mathbb{P}^*$*?*

Problem 18.3: *Consider a non-dividend-paying stock and European call option on that stock. As usual, let K denote the option's strike price and let T denote its exercise date.* Under the risk-neutral probability measure $\mathbb{P}^*$, *what is the probability that the call is* ***in-the money*** *at time-T?*

Numerical Exercises.

Problem 18.4: *Assume the Black-Scholes model. The current stock price is* 50 *and the stock's volatility is* 0.25*. The stock pays no dividends. The continuously compounded risk-free interest rate is* 0.05*. Consider a European call option on this stock with strike price* 55 *and exercise date in six months.* Under the risk-neutral probability measure $\mathbb{P}^*$*, what is the probability that the call is* ***in-the-money*** *at expiration?*

Problem 18.5: *Assume the Black-Scholes model. The current stock price is* 80 *and the stock's volatility is* 0.30*. The stock pays no dividends. The continuously compounded risk-free interest rate is* 0.05*. Consider a one-year,* 75*-strike European put option on this stock.* Under the risk-neutral probability measure $\mathbb{P}^*$*, what is the probability that the put is* ***in-the-money*** *at expiration?*